Experiment No. 10

Title:

Exploring Linux Inodes, I/O Redirection, Piping, and Process Control.

Aim:

To understand the concept of inodes in Linux, practice input/output (I/O) redirection and piping, and use process control commands to manage running processes in a Linux environment.

Theory:

Inodes

An inode is a data structure that stores metadata of a file such as:

- File size
- Ownership
- Permissions
- Timestamps
- Disk location

File names are stored separately in directory entries and point to inodes.

I/O Redirection

Redirection allows changing the standard input, output, or error streams:

- > – Redirect output (overwrite)
- >> – Redirect output (append)
- < – Redirect input
- 2> – Redirect error output subdirectories.

Pipes

Pipes (|) allow the output of one command to be used as input to another command.

Example:

- ls | grep txt

Process Control

Linux supports multitasking. Processes can be monitored and controlled using commands such as:

- ps
- top
- kill
- jobs
- bg, fg

Commands Used

Commands	Description
ls -l	Display inode number of files
stat	Display file inode details
>, >>	Output redirection
<	Input redirection
ps	Display running processes
top	Real-time process monitoring
kill	Terminate a process
jobs	List background jobs
bg	Resume job in background

Procedure / Steps

A. Understanding Inodes:

Step 1: Login

- Log in to the Linux virtual machine and open the **Terminal**.

Step 2: Create a test file using:

- touch inode_test.txt

```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal]
└─# touch inode_test.txt

┌──(root㉿kali)-[/home/aman_singhal]
└─#
```

Step 3: Display inode number using:

- `ls -i inode_test.txt`

```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal]
└─# ls -i inode_test.txt
4335394 inode_test.txt

┌──(root㉿kali)-[/home/aman_singhal]
└─#
```

Step 4: View inode details:

- `stat inode_test.txt`

```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal]
└─# stat inode_test.txt
  File: inode_test.txt
  Size: 0               Blocks: 0          IO Block: 4096   regular empty fil
e
Device: 8,1     Inode: 4335394     Links: 1
Access: (0644/-rw-r--r--)  Uid: (    0/    root)   Gid: (    0/    root)
Access: 2026-03-17 02:08:09.212236427 -0400
Modify: 2026-03-17 02:08:09.212236427 -0400
Change: 2026-03-17 02:08:09.212236427 -0400
 Birth: 2026-03-17 02:08:09.212236427 -0400

┌──(root㉿kali)-[/home/aman_singhal]
└─#
```

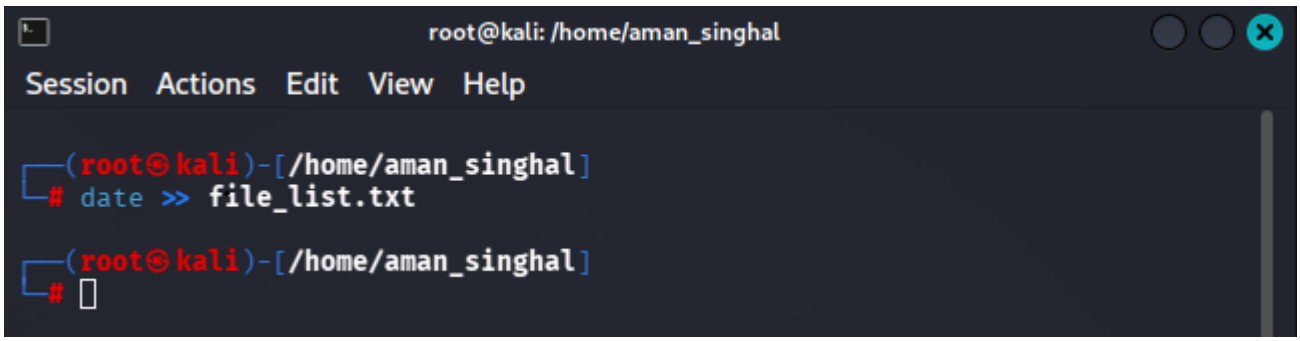
B. Input / Output Redirection:

Step 5: Redirect output to a file using:

- ls > file_list.txt

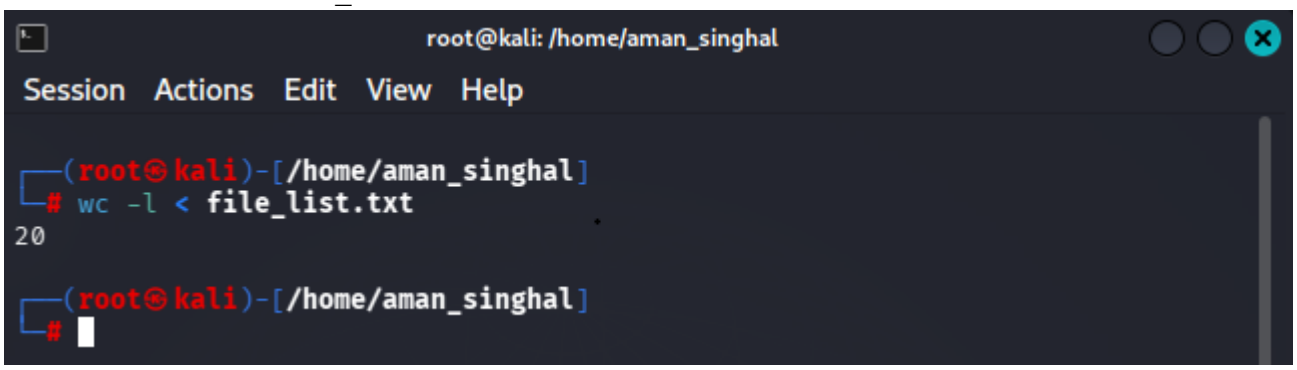
Step 6: Append output Using:

- date >> file_list.txt



Step 7: Redirect input Using:

- wc -l < file_list.txt



C. Using Pipes:

Step 8: Combine commands using pipe Using:

- `ls -l | grep ".txt"`

```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal]
└─# ls -l | grep ".txt"
-rw-r--r-- 1 root root   0 Feb 28 10:16 file1.txt
-rw-r--r-- 1 root root   0 Feb 28 10:16 file2.txt
-rw-r--r-- 1 root root 211 Mar 17 02:10 file_list.txt
-rw-r--r-- 1 root root   0 Mar 17 02:08 inode_test.txt

┌──(root㉿kali)-[/home/aman_singhal]
└─#
```

Step 9: Count number of text files Using:

- `ls | grep ".txt" | wc -l`

```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal]
└─# ls | grep ".txt" | wc -l
4

┌──(root㉿kali)-[/home/aman_singhal]
└─#
```

D. Process Control Commands:

Step 10: Display running processes using:

- `ps`

```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal]
└─# ps
    PID TTY          TIME CMD
   8486 pts/1    00:00:00 sudo
   8487 pts/1    00:00:00 su
   8489 pts/1    00:00:02 zsh
  11678 pts/1    00:00:00 top
  20555 pts/1    00:00:00 ps

┌──(root㉿kali)-[/home/aman_singhal]
└─#
```

Step 11: Monitor processes in real-time Using:

- `top`

```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help
top - 02:13:00 up 29 min,  1 user,  load average: 0.15, 0.07, 0.06
Tasks: 263 total,   1 running, 261 sleeping,   1 stopped,   0 zombie
%Cpu(s):  0.3 us,  0.5 sy,  0.0 ni, 99.1 id,  0.0 wa,  0.0 hi,  0.1 si,  0.0
MiB Mem :   9850.0 total,   7414.8 free,   1009.5 used,   1683.8 buff/cache
MiB Swap:    953.7 total,    953.7 free,      0.0 used.   8840.5 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+
    909 root      20   0  456848 147720  89136 S   1.0   1.5   0:33.78
   1701 aman_si+  20   0 1499872 143408 100200 S   0.7   1.4   0:05.90
    703 root      20   0  352936   2784   2516 S   0.3   0.0   0:02.49
   1473 aman_si+  20   0  214900   3312   2856 S   0.3   0.0   0:01.94
   1482 aman_si+  20   0  280952   3292   2876 S   0.3   0.0   0:06.57
   1497 aman_si+  20   0  215088   3564   3084 S   0.3   0.0   0:00.93
   1831 aman_si+  20   0  329716  81436  24764 S   0.3   0.8   0:06.28
   1835 aman_si+  20   0  277088  30228  22912 S   0.3   0.3   0:05.48
   1843 aman_si+  20   0 1024056  38244  30052 S   0.3   0.4   0:01.03
   1930 root      20   0  318780  11016   9060 S   0.3   0.1   0:01.54
   8366 aman_si+  20   0  808440  67528  50552 S   0.3   0.7   0:02.51
  11071 root      20   0       0      0      0 I   0.3   0.0   0:00.03
  19989 aman_si+  20   0  212656   6760   6088 S   0.3   0.1   0:00.02
  19991 aman_si+  20   0  212656   6888   6224 S   0.3   0.1   0:00.02
  20738 root      20   0   10420   5936   3792 R   0.3   0.1   0:00.02
      1 root      20   0   24724  15492  11480 S   0.0   0.2   0:01.82
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.08
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00
```

Step 12: Run a command in background Using:

- `sleep 60 &`

```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal]
└─# sleep 60 &
[3] 21025

┌──(root㉿kali)-[/home/aman_singhal]
└─#
```

Step 13: List background jobs Using:

- `jobs`

```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal]
└─# jobs
[1]  - suspended (signal)  top
[2]  + suspended (signal)  top
[3]    running    sleep 60

┌──(root㉿kali)-[/home/aman_singhal]
└─#
```

Step 14: Bring job to foreground Using:

- `fg`

```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help
top - 02:13:48 up 30 min,  1 user,  load average: 0.06, 0.05, 0.06
Tasks: 264 total,   1 running, 262 sleeping,   1 stopped,   0 zombie
%Cpu(s):  0.5 us,  0.6 sy,  0.0 ni, 98.8 id,  0.0 wa,  0.0 hi,  0.2 si,  0.0
MiB Mem :   9850.0 total,   7412.7 free,   1010.7 used,   1684.6 buff/cache
MiB Swap:    953.7 total,    953.7 free,      0.0 used.   8839.2 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+
    909 root      20   0  456848 147720  89136 S   1.7   1.5   0:34.65
   8489 root      20   0   10588   6876   4612 S   0.6   0.1   0:02.68
   1701 aman_si+  20   0 1499872 143408 100200 S   0.4   1.4   0:06.08
   1831 aman_si+  20   0  329716  81436  24764 S   0.3   0.8   0:06.44
   1482 aman_si+  20   0  280952   3292   2876 S   0.3   0.0   0:06.71
   1921 aman_si+  20   0  695380  37060  28580 S   0.3   0.4   0:01.20
   8366 aman_si+  20   0  808440  67536  50552 S   0.3   0.7   0:02.66
   1835 aman_si+  20   0  277088  30228  22912 S   0.3   0.3   0:05.61
    703 root      20   0  352936   2784   2516 S   0.1   0.0   0:02.55
   2213 aman_si+  20   0  624272  41240  23996 S   0.1   0.4   0:00.89
   1258 aman_si+  20   0  832556  42428  24012 S   0.1   0.4   0:01.11
   1473 aman_si+  20   0  214900   3312   2856 S   0.1   0.0   0:01.99
   1771 aman_si+  20   0 1455740  68056  46212 S   0.1   0.7   0:02.02
     15 root      20   0       0      0      0 I   0.1   0.0   0:01.38
     90 root      20   0       0      0      0 I   0.1   0.0   0:01.65
   1754 aman_si+  20   0 1178904  44756  32248 S   0.1   0.4   0:01.38
   1930 root      20   0  318780  11016   9060 S   0.1   0.1   0:01.58
  20326 aman_si+  20   0  212656   8252   6148 S   0.1   0.1   0:00.10
    978 systemd+  20   0   16668   7564   6504 S   0.1   0.1   0:00.92
   1239 aman_si+  20   0    9092   6148   4324 S   0.1   0.1   0:00.56
   1790 aman_si+  20   0  169080   7224   6408 S   0.1   0.1   0:00.39
```

Step 15: Terminate a process Using:

- `Kill <PID>`

Outcomes

- Explain the concept of inodes and identify inode numbers of files in a Linux filesystem.
- Apply input and output redirection to store command output or provide input to commands from files.
- Utilize piping to chain multiple commands for efficient data processing.
- Monitor and manage Linux processes using process control commands.
- Demonstrate the practical use of these Linux features in automation, scripting, and system administration tasks.

Result

- Each file in Linux is associated with a unique inode containing metadata such as file type, permissions, owner, size, and timestamps.
- File content can be redirected to and from files using input (<) and output (> or >>) redirection operators.
- Piping (|) allows the output of one command to be used as input to another command, enabling efficient data processing.
- Process control commands such as ps, top, kill, fg, bg, and jobs allow monitoring, suspending, resuming, and terminating processes.
- Combined usage of these mechanisms facilitates automation, scripting, and efficient system resource management.